Jinho Chang

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Research Profile

Ph.D. student at the Graduate School of A.I., KAIST. I am working on diffusion and flow-matching generative models for their variational applications and inference-time techniques, and multi-modal models for informative representations and cross-modal generation.
Also, leveraging my chemistry background, I aim to apply state-of-the-art AI techniques to various important chemical challenges with minimal adaptation costs.

Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*

Ph.D. in Artificial Intelligence (M.S./Ph.D. integrated program) Expected Aug. 2028

M.S. stage: Sep. 2022–Aug. 2024; Ph.D. stage: Sep. 2024–present

B.S. in Computer Science and Chemistry (double major) Mar. 2018–Aug. 2022

Research Areas

Generative Modeling & Alignment: Diffusion/flow matching, reward-based fine-tuning, training-free guidance, stochastic optimal control, and image/video editing.

Molecular AI & Multimodal Learning: Text-conditioned molecule generation, molecular foundation models, molecular property prediction, and cross-modal representations.

Selected Publications

Full publication list: Google Scholar. *Equal contribution.

Molecular AI & Drug Discovery

LDMol: Text-Conditioned Molecule Diffusion Model Leveraging Chemically Informative Latent Space [Paper]

Jinho Chang, Jong Chul Ye *ICML 2025*

Introduces text-conditioned molecular diffusion with a chemically informative contrastive latent space; supports generation, retrieval, and molecular editing.

Bidirectional Generation of Structure and Properties Through a Single Molecular Foundation Model [Paper]

Jinho Chang, Jong Chul Ye *Nature Communications 2024*

Unifies molecular structure and biochemical property representations for conditional generation, property prediction, and related molecular tasks.

Drug-likeness Scoring Based on Unsupervised Learning [Paper]

Jinho Chang, Kyunghoon Lee, Seonghwan Seo, Woo Youn Kim *Chemical Science 2022*

Develops an unsupervised language-model-based scoring method for assessing molecular drug-likeness.

Generative Model & Alignment

Manifold-Constrained Initial Noise Optimization for Efficient Generative Model Alignment [Paper]

Jinho Chang, Jong Chul Ye *Under Review, 2026*

Optimizes diffusion/flow initial noise for black-box rewards using soft-value guidance and manifold-constrained Langevin and Metropolis–Hastings updates.

Reward Score Matching: Unifying Reward-based Fine-tuning for Flow and Diffusion Models [Paper]

Jeongjae Lee*, **Jinho Chang***, Jeongsol Kim, Jong Chul Ye *NeurIPS 2026; ICML2026 SPIGM Workshop Best Paper*

Unifies reward-aligned fine-tuning as value-guided score matching, clarifying estimation and optimization tradeoffs for differentiable and black-box rewards.

Contrastive CFG: Improving CFG in Diffusion Models by Contrasting Positive and Negative Concepts [Paper]

Jinho Chang, Changsun Lee, Hyungjin Chung, Jong Chul Ye *ICML 2026*

Introduces contrastive guidance for negative concepts, improving unwanted-feature removal while preserving image quality.

Training-Free Reward-Guided Image Editing via Trajectory Optimal Control [Paper]

Jinho Chang[*], Jaemin Kim[*], Jong Chul Ye *ICLR 2026*

Formulates reward-guided editing as trajectory optimal control, improving reward without sacrificing source-image fidelity.

DreamMotion: Space-Time Self-Similarity Score Distillation for Zero-Shot Video Editing [Paper]

Hyeonho Jeong, **Jinho Chang**, Geon Yeong Park, Jong Chul Ye *ECCV 2024*

Uses space-time self-similarity score distillation for video editing while preserving motion and scene structure.

LLM-CXR: Instruction-Finetuned LLM for CXR Image Understanding and Generation [Paper]

Suhyeon Lee[*], Wonjun Kim[*], **Jinho Chang**, Jong Chul Ye *ICLR 2024*

Instruction-tunes a language model to understand and generate chest X-ray images using visual tokens.

Research & Professional Experience

Research Intern, HITS Jan. 2021–Jun. 2021

Developed a reinforcement learning algorithm on a scaffold-based molecule generation model for synthesizable molecules with targeted pharmaceutical properties.

Lab Manager, BioImaging-Signal Processing & Learning Lab (BISPL), KAIST 2026

Coordinated laboratory operations, research proposals, resources, and team members.

Technical Skills

Programming & frameworks: Python, PyTorch, RDKit, NumPy & Pandas.
Methodology: Diffusion/flow models, representation learning, inference-time control, reinforcement learning, optimal control, molecular generative modeling.

Awards & Honors

Best Paper Award, ICML 2026 SPIGM Workshop (Reward Score Matching) 2026
Top Reviewer Award, ICML 2026(Gold) 2026
National Science & Technology Scholarship, Ministry of Science and ICT, South Korea 2019–2022
KAIST Presidential Fellowship (KPF) 2018–2022

Academic Service

Journal reviewer: Nature Communications; Communications Chemistry (2024–present).
Conference reviewer: ICLR (2025–2027); CVPR (2025); ICCV (2025); ICML (2026); NeurIPS (2026).